Venkata Lakshmi Parimala Pasupuleti

Risk Analysis Lead · Multifamily Affordable Housing · Counterparty Risk · Fraud Risk · Compliance · LIHTC · Section 8 · HCV · HUD · Python · SQL · Tableau

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PROFESSIONAL SUMMARY

Risk Analysis Lead with 8+ years in multifamily affordable housing compliance, **counterparty risk management**, fraud risk detection, and technology/data controls across **Section 8/HCV, LIHTC, and HUD** portfolios — the same federal framework underpinning Fannie Mae's DUS lender ecosystem. **Designed and implemented operational risk controls and KRI monitoring** covering 100K+ units with zero compliance violations: AMI threshold calculations, income eligibility verification, recertification governance, escalation routing, and **cybersecurity/data controls** (100% PHI compliance, Libsodium encryption, zero security incidents). Eliminated **80% of manual operational workload** through AI-enabled compliance automation; built production AI pipeline (1.6M financial records) and agentic audit system achieving **93.6% precision across 10 compliance rules**. Built quantitative models achieving **$R^2 = 0.977$**; dashboards improving decision-making comprehension **40%** (Tableau, Power BI). M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026).

WORK EXPERIENCE

Entrata

Jan 2017 – Aug 2024

Senior Software Engineer — Multifamily Risk Management · Affordable Housing Compliance · Financial Analytics

- **Designed and implemented counterparty risk controls, risk monitoring framework, and compliance monitoring** for Section 8/HCV, LIHTC, and HUD multifamily portfolios — AMI threshold KRI monitoring, **fraud risk management** (income misrepresentation, fake paystubs, household composition fraud), recertification governance, escalation routing, and control gap identification across 100K+ unit portfolios; **zero compliance violations** over 7+ years; led **due diligence** reviews for counterparty onboarding, coordinating cross-functionally across compliance, operations, legal, and product stakeholders to assess operational readiness and remediate risks.
- **Led financial analysis, risk framework development, and governance approval processes** for affordable housing programs: HAP allocation, AMI-based pricing tiers, tax credit calculations, HUD payments, rental subsidies, and ledger management — assessed lender/servicer compliance with federal housing requirements; evaluated technology and **data controls** including cybersecurity (Libsodium encryption, PHI pipeline) achieving **100% data compliance** with zero security incidents; built **third-party risk integrations** (Salesforce CRM data pipelines, API connections) ensuring vendor operational readiness.
- Built **AI-enabled solutions** processing 1.6M financial records (Python, PyTorch, OpenAI API) — automated compliance workflows, output validation, quality gates; eliminated **80% of manual operational workload**; optimized **200+ SQL queries** (30% performance gain); drove **project management** across 120+ Agile sprints; mentored 5+ engineers across 300+ code reviews.

George Mason University

Aug 2024 – Aug 2025

Research Associate — Quantitative Risk Analytics · Financial Modeling · Stakeholder Reporting

- Built multi-source quantitative financial models (Python, R) achieving **$R^2 = 0.977$** across 190+ countries — data collection, hypothesis testing, trend identification, and validation pipeline; delivered risk analysis and strategy recommendations to **senior leadership** stakeholders.
- Surfaced financial risk anomalies manual review missed across 3 research pipelines by applying **Isolation Forest** — identified control gaps and proposed corrective actions for downstream model reliability.
- Designed strategic risk dashboards (Tableau, ggplot2, Quarto) improving decision-making comprehension **40%** across two cohorts; mentored **600+ students** in data analysis, Python, and database design as Graduate Teaching Assistant.

PROJECTS

Leasel: AI Compliance Audit Pipeline for Affordable Housing — 93.6% Precision

Python · LangGraph · Scikit-learn · SQLite · Deterministic Rules Engine · reportlab — github.com/venkatalakshmiparimala/Lease_complaince_rag

- Built an end-to-end AI compliance audit pipeline for affordable housing lease documents (1,240 files, 5 US states + federal standards, LIHTC/Section 42) — deterministic rule-based engine with full data lineage; achieved **93.6% precision, 87.9% F1** across 10 compliance rules; every determination traceable and audit-ready. Directly mirrors Fannie Mae's compliance monitoring and control framework.

CrashClarity: Quantitative Risk Analytics on 104K Records

Python · R · Scikit-learn · Random Forest · Logistic Regression · Feature Engineering · Tableau — Live Dashboard

- Mined 104K+ records — built Random Forest + Logistic Regression risk classification model with hypothesis testing achieving **85%+ accuracy**; deployed self-service Tableau dashboard translating quantitative risk outputs into actionable insights for non-technical senior stakeholders.

EDUCATION

M.S. Data Analytics Engineering — George Mason University — Graduated May 2026

GPA: 3.87 · Fairfax, Virginia · Quantitative risk modeling, financial analytics, compliance analytics, anomaly detection, data governance, stakeholder reporting

B.Tech Computer Science & Engineering — Jawaharlal Nehru Technological University, Anantapur — 2013 – 2017

TECHNICAL SKILLS

Risk Management & Compliance: Counterparty Risk Assessment, Fraud Risk Management, Operational Risk Controls, Cybersecurity & Technology/Data Controls, Third-Party Risk, KRI Monitoring, Control Gap Identification, Operational Readiness Assessment, Risk Mitigation Strategy, Compliance Testing, Multifamily Affordable Housing (LIHTC, Section 8/HCV, HUD, AMI), Recertification Governance, Escalation Routing, PHI/HIPAA Compliance, Audit-Ready Pipelines, Governance & Approval Processes

Analytics & Modeling: Python (expert), R, SQL (expert — 200+ production optimizations), PL/SQL; Quantitative Risk Modeling, Hypothesis Testing, Regression Analysis, Isolation Forest (Anomaly Detection), Random Forest, Logistic Regression, Feature Engineering, Scikit-learn, Pandas, NumPy; Tableau, Power BI, ggplot2, Quarto, Excel (advanced)

AI & Cloud Infrastructure: AI in financial workflows (LangGraph, RAG pipelines, OpenAI API, PyTorch), AWS (S3, EC2, RDS, Glue), ETL/ELT Pipelines, PostgreSQL, MySQL, Docker, GitHub Actions, Salesforce (CRM integrations), Git, Agile/Scrum (120+ sprints)